

Fine-Tuning can Distort Pretrained Features and Underperform Out-of-Distribution

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Ananya Kumar et al.

Presenter

이재윤

Fundamental Team

김채현, 양현모, 이근배

Contents

1. Introduction

2. Theory

3. Experiments

4. Conclusion

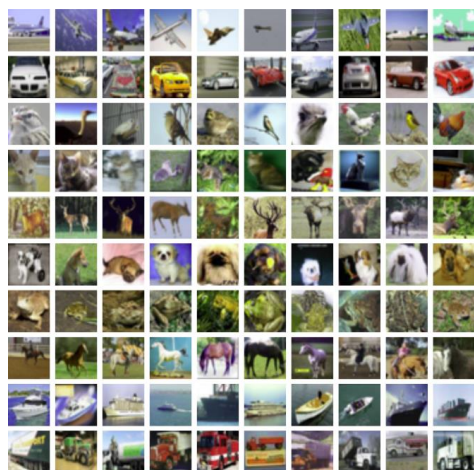
1. Introduction

Introduction

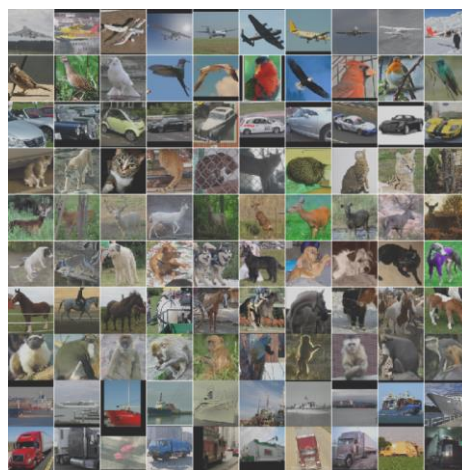
Fine-Tuning can Distort Pretrained Features and
Underperform ***Out-of-Distribution***

Out-of-Distribution(OOD)

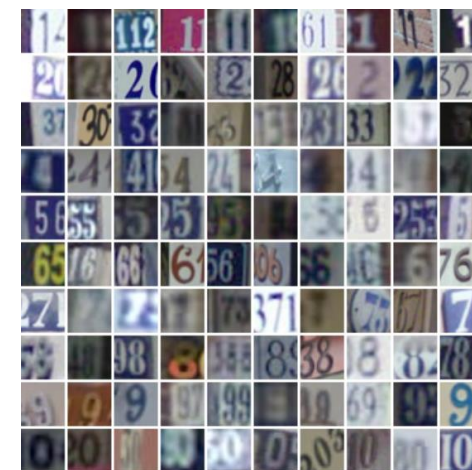
- Term **OOD** in this paper is different from **OOD** in Out-of-Distribution detection
 - Out-of-Distribution in this paper means data distribution that was not seen during training
 - Assume ImageNet pretrained network is fine-tuned by **CIFAR-10 dataset**
 - ✓ **CIFAR-10** : In-Distribution(ID)
 - ✓ **STL** : Out-of-Distribution(OOD)
- *The class categories of CIFAR-10 and STL are the same



<CIFAR-10>



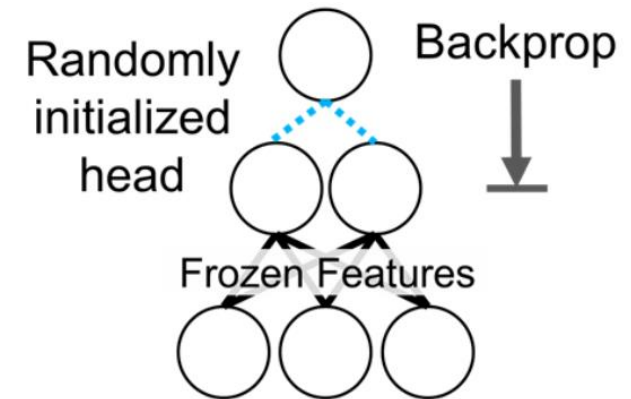
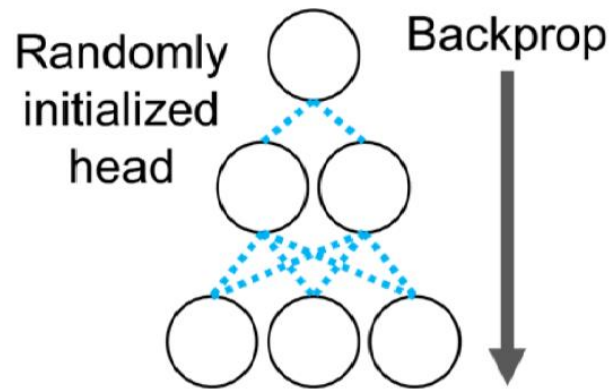
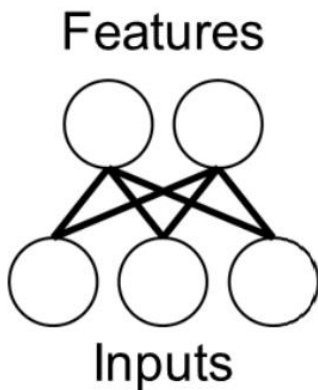
<STL>



<SVHN>

Transfer Learning

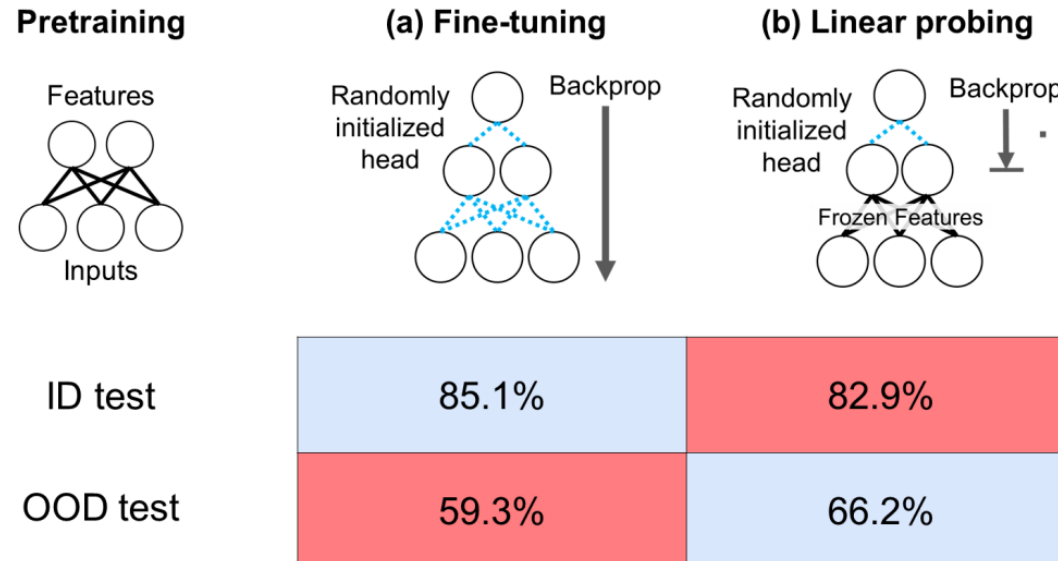
- **Transfer learning (TL)** is a research problem in machine learning (ML) that focuses on storing knowledge gained while solving one problem and **applying it to a different but related problem**^[1]
- Two main techniques are known for transfer learning
 - I. **Fine-Tuning**
 - ✓ Updates all parameters of model
 - II. **Linear Probing**
 - ✓ Updates only the parameters of the last linear layer



[1] https://en.wikipedia.org/wiki/Transfer_learning

Fine-Tuning vs Linear Probing

- It is well known that fine-tuning has higher accuracy than linear probing
- **In certain circumstances**, however, fine-tuning may show **lower accuracy** than linear probing
 - ✓ When?
 - Distribution shift between ID(In-Distribution) and OOD(Out-of-Distribution) is large
 - ✓ Why?
 - **Fine-Tuning** distorts pretrained features



Question

2. Theory

Setup

- Learn a **predictor** $f : \mathbb{R}^d \rightarrow \mathcal{Y}$ to map inputs $x \in \mathbb{R}^d$ to outputs $y \in \mathcal{Y}$

- Evaluate both on In-Distribution and Out-of-Distribution

$$\mathcal{L}_{id} = \mathbb{E}_{(x,y) \sim P_{id}}[\ell(f(x), y)] \text{ and } \mathcal{L}_{ood} = \mathbb{E}_{(x,y) \sim P_{ood}}[\ell(f(x), y)]$$

- Predictor is parameterized as follows:

$$f_{v,B}(x) = v^T g_B(x) ,$$

where $g_B(x) \in \mathbb{R}^k$ is a feature, B is feature extractor and v is **linear head**

- Assume the pretrained **feature extractor** B_0 is obtained by training on large amounts of data
 - ✓ **Linear Probing** minimize the loss maintaining the initial feature extractor $B = B_0$
 - ✓ **Fine-Tuning** minimize the loss by updating both feature extractor B and linear head v

Toy Experiment Setup

- For analysis, paper focus on **regression** task

$$\mathcal{Y} = \mathbb{R} \text{ and } \ell(\hat{y}, y) = (\hat{y} - y)^2$$

- Assume the linear feature extractor, a model consists of two linear layered network

$$f_{v,B}(x) = v^T Bx$$

where $B \in \mathcal{B} = \mathbb{R}^{k \times d}$, and $v \in \mathcal{V} = \mathbb{R}^k$

- Good pretrained model

- ✓ For simplicity, models are assumed to be well-specified i.e. $y = v_*^T B_* x$
- ✓ For any rotation matrix U , $(Uv_*)^T (UB_*)x = v_*^T B_* x$
- ✓ Suppose we have a pretrained feature extractor B_0 , close to B_* , so $d(B_0, B_*) \leq \epsilon$ where the distance d is defined below:

$$d(B, B') = \min_U \|B - UB'\|_2.$$

- ✓ Pretraining is done on large scale data so that B_0 becomes good pretrained model

Toy Experiment Setup

- Training Data
 - ✓ Let $X \in \mathbb{R}^{n \times d}$, $X \neq 0$ be a matrix of n training examples from P_{id} .
 - ✓ Let $Y \in \mathbb{R}^n$ be the corresponding outputs.
- Let $S = \text{rowspace}(X)$ be the m -dimensional subspace spanning the training examples.
 - ✓ This paper considers an overparameterized setting
$$1 \leq m < d - k \quad (d = 10K, k = 100, m = 5K).$$
- OOD data is assumed to contains examples outside the span of the training data
 - ✓ Let P_{ood} have the second moment $\Sigma = \mathbb{E}[xx^T]$ where $x \sim P_{ood}$

Training Method

- Fine-Tuning and Linear Probing both optimize the loss via gradient descent
- Formally, gradient flow differential equations for LP and FT are as follows:

$$\partial_t \mathbf{v}_{ft}(t) = -\nabla_v \hat{L}(\mathbf{v}_{ft}(t), \mathbf{B}_{ft}(t)), \partial_t \mathbf{B}_{ft}(t) = -\nabla_B \hat{L}(\mathbf{v}_{ft}(t), \mathbf{B}_{ft}(t)),$$

$$\partial_t \mathbf{v}_{lp}(t) = -\nabla_v \hat{L}(\mathbf{v}_{lp}(t), \mathbf{B}_0), \partial_t \mathbf{B}_{lp}(t) = \mathbf{0}$$

Initialized with $\mathbf{B}_{ft}(0) = \mathbf{B}_{lp}(0) = \mathbf{B}_0$, and $\mathbf{v}_{ft}(0) = \mathbf{v}_{lp}(0) = \mathbf{v}_0$

- The final LP and FT solutions are the limit points of the corresponding gradient flows:

$$\mathbf{v}_{ft}^\infty = \lim_{t \rightarrow \infty} \mathbf{v}_{ft}(t) \text{ and } \mathbf{B}_{ft}^\infty = \lim_{t \rightarrow \infty} \mathbf{B}_{ft}(t)$$

$$\mathbf{v}_{lp}^\infty = \lim_{t \rightarrow \infty} \mathbf{v}_{lp}(t) \text{ and } \mathbf{B}_{lp}^\infty = \lim_{t \rightarrow \infty} \mathbf{B}_{lp}(t) = \mathbf{B}_0$$

When and Why the distortion happens?

- Two main observation

- ① Features get distorted

- ✓ Representation only change only in the ID subspace
- ✓ Representation are unchanged in the orthogonal subspace
- ✓ Derivatives of the training loss $\hat{L}(v, B) = \|XB^T v - Y\|_2^2$ with respect to B

$$\nabla_B \hat{L}(v, B) = 2v(Y - XB^T v)^T X$$

- ✓ Gradients updates to B do not modify Bu for $u \in S^\perp$ ($S = \text{rowspace}(X)$)

$$\nabla_B \hat{L}(v, B)u = 2v(Y - XB^T v)^T Xu = \mathbf{0}$$

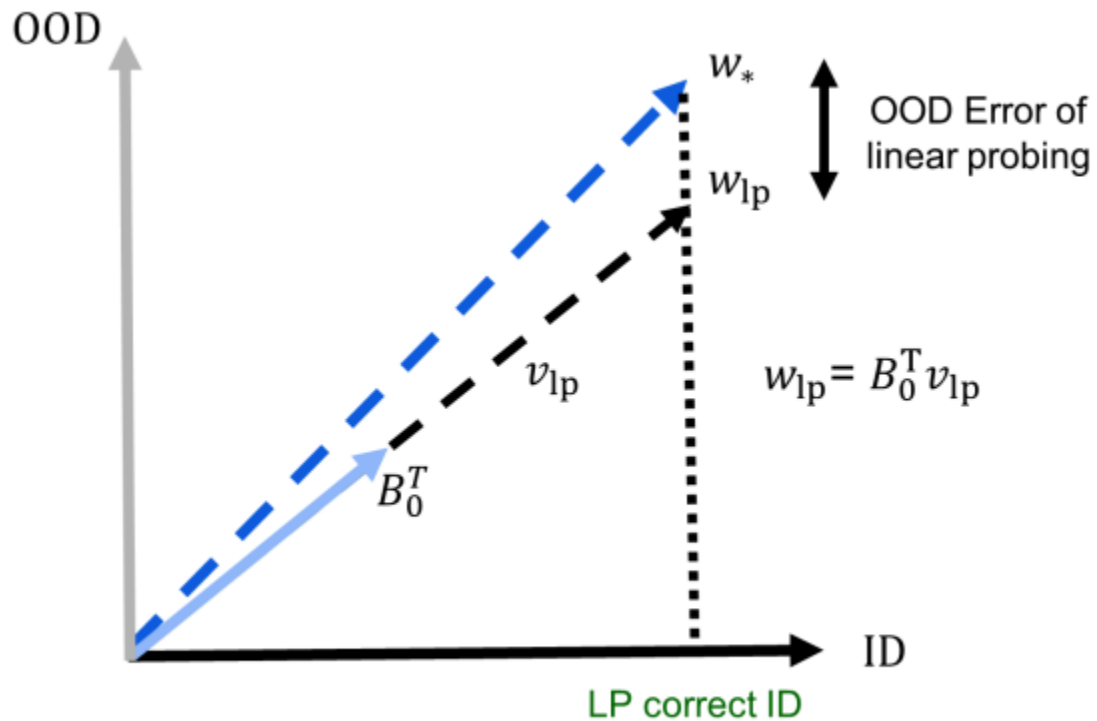
- ✓ However, **gradient is non-zero for directions u in the ID subspace** and the corresponding features Bu



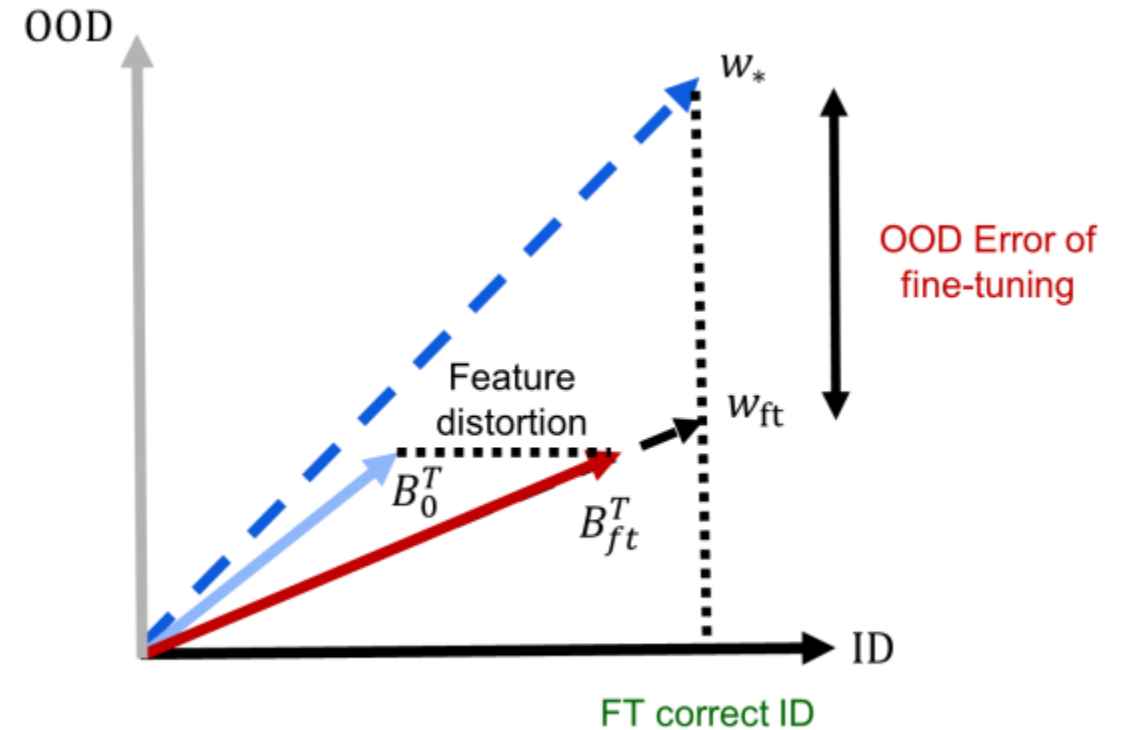
Feature distortion

When and Why the distortion happens?

- ② Distorted features can lead to higher OOD error
- ✓ Let's consider toy example where $d = 2$ and $k = 1$
 - ✓ Linear head v is a scalar quantity how much the feature extractor B be scaled



(a) Toy example (Linear probing)



(b) Toy example (fine-tuning)

OOD error Lower bound

- For all time steps t , the OOD error of the FT iterates is *lower bounded*:

$$\sqrt{L_{ood}(v_{ft}(t), B_{ft}(t))} > \sqrt{\sigma_{min}(\Sigma)} \left(\frac{\cos \theta_{max}(R_0, S^\perp)}{\sqrt{k}} \frac{\min\left(\varphi, \frac{\varphi^2}{\|w_*\|_2}\right)}{(1 + \|W_*\|_2)^2} + \epsilon \right)$$

where $\varphi^2 = |(v_0^T v_*)^2 - (v_*^T v_*)^2|$ and $\epsilon \geq d(B_0, B_*)$

$$L_{ood}(v, B) = \mathbb{E}[(v_*^T B_* x - v^T B x)^2] = (B_*^T v_* - B^T v)^T \Sigma (B_*^T v_* - B^T v)$$

B : feature extractor
 v : linear head
 R_0 : rowspace of B_0
 S : rowspace of X
 $\cos \theta_{max}(A, B)$: Largest principle angle of subspace A and B
 φ : head alignment error
 ϵ : error in the pretrained feature extractor
 v_*, B_* : optimal parameters

$$\|B_*^T v_* - B_{ft}^T v_{ft}^t\|_2^2 \geq \frac{c}{\sqrt{k}} \frac{\min\left(\frac{\varphi}{\|w_*\|_2}, \varphi^2\right)}{(1 + \|w_*\|_2)^2} - \epsilon$$

Interpretations of Various Quantities

- For all time steps t , the OOD error of the FT iterates is *lower bounded*:

$$\sqrt{L_{ood}(v_{ft}(t), B_{ft}(t))} > \sqrt{\sigma_{\min}(\Sigma)} \left(\frac{\mathbf{c}}{\sqrt{k}} \boldsymbol{\varphi} + \boldsymbol{\epsilon} \right)$$

where $\mathbf{c} = \cos \theta_{\max}(R_0, \mathbf{S}^\perp)$ and if $\|w\|_2 = 1$ and $\boldsymbol{\varphi}$ is bounded by some constant

I. Alignment error of random head initialization($\boldsymbol{\varphi}$)

- ✓ Lower bound increases as $\boldsymbol{\varphi}$ increases
- ✓ Alignment error increases because the gradient updates to the v, B are coupled
- ✓ If the head is somehow initialized perfectly at v_* , fine-tuning **may not increase OOD error**

II. Span of Training data($\mathbf{S}, \mathbf{c}$)

- ✓ If the training dataset is small, ID distribution may not be spanned by the training data
- ✓ ID error and OOD error increase

III. Quality of pretrained features($\boldsymbol{\epsilon} \geq d(B, B') = \min_U \|B - UB'\|_2$)

- ✓ If pretrained features are perfect ($\epsilon = 0$) lower bound becomes 0
- ✓ Worse features, lower bound increases

Linear Probing then Fine-Tuning (LP-FT)

- Fine-tuning can adapt both feature extractor B and linear head v to fit downstream task
 - ✓ Low OOD accuracy due to the feature distortion
- By aforementioned theorem, alignment error φ plays an important role
 - ✓ With FT, under random or zero initialization, φ^2 is usually large
 - ✓ Better head initialization is needed – one obtained by LP
- If the pretrained features are descent, a linear probed head would be much better aligned with v_*

Question

3. Experiments

Experiments settings

- Datasets / Architecture

ID	DomainNet	Living-17	FMoW Geo-Shift (North America)	CIFAR-10	CIFAR-10	ImageNet-1K
OOD	DomainNet	Entity-30	FMoW Geo-Shift (Africa ,Europe)	STL	CIFAR-10.1	ImageNet V2 ImageNet-R ImageNet-A ImageNet-Sketch
Architecture	ResNet-50	ResNet-50	ResNet-50	ResNet-50	ResNet-50	CLIP pretrained ViT-B/16

- Training method

- I. Fine-Tuning

- ✓ Cosine learning rate schedule
- ✓ 64 batch size
- ✓ Early stop and choose the best learning rate using ID validation dataset

- II. Linear-Probing

- ✓ Train l_2 -regularized logistic regression classifier
- ✓ Select the best l_2 -regularization hyperparameter using ID validation accuracy

Results - LP vs FT vs LP-FT

- Fine-Tuning does better than Linear Probing on 5/6 ID datasets
- Linear-Probing does better than Fine-Tuning on 8/10 OOD datasets
- Results consistent with prior work and intuitions to some content
 - ✓ Exception in case of CIFAR-10.1 and ImageNetV2
 - ✓ There is **no such large distributional shift** between ID and OOD data (CIFAR-10 ↔ CIFAR-10.1 / ImageNet ↔ ImageNetV2)

	CIFAR-10	Ent-30	Liv-17	DomainNet	FMoW	ImageNet	Average
FT	97.3 (0.2)	93.6 (0.2)	97.1 (0.2)	84.5 (0.6)	56.5 (0.3)	81.7 (-)	85.1
LP	91.8 (0.0)	90.6 (0.2)	96.5 (0.2)	89.4 (0.1)	49.1 (0.0)	79.7 (-)	82.9
LP-FT	97.5 (0.1)	93.7 (0.1)	97.8 (0.2)	91.6 (0.0)	51.8 (0.2)	81.7 (-)	85.7

ID Accuracy

	STL	CIFAR-10.1	Ent-30	Liv-17	DomainNet	FMoW
FT	82.4 (0.4)	92.3 (0.4)	60.7 (0.2)	77.8 (0.7)	55.5 (2.2)	32.0 (3.5)
LP	85.1 (0.2)	82.7 (0.2)	63.2 (1.3)	82.2 (0.2)	79.7 (0.6)	36.6 (0.0)
LP-FT	90.7 (0.3)	93.5 (0.1)	62.3 (0.9)	82.6 (0.3)	80.7 (0.9)	36.8 (1.3)

OOD Accuracy

	ImNetV2	ImNet-R	ImNet-Sk	ImNet-A	Average
FT	71.5 (-)	52.4 (-)	40.5 (-)	27.8 (-)	59.3
LP	69.7 (-)	70.6 (-)	46.4 (-)	45.7 (-)	66.2
LP-FT	71.6 (-)	72.9 (-)	48.4 (-)	49.1 (-)	68.9

**OOD (ImageNet)
Accuracy**

Results - LP vs FT vs LP-FT

- LP-FT use similar compute compared to fine-tuning
- LP-FT follows the same training setting of fine-tuning
- LP-FT gets the best accuracy ID and OOD, true for 5/6 ID and 10/10 OOD datasets
 - ✓ Except for ID accuracy on FMoW
 - ✓ This could be because the pretrained features are not good

	CIFAR-10	Ent-30	Liv-17	DomainNet	FMoW	ImageNet	Average
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OOD (ImageNet) Accuracy

Examining the feature distortion theory

1. Early stopping does not mitigate feature distortion
 - ✓ One might think fine-tuning simply overfitting ID, and so early stopping on OOD might match linear probing OOD
 - ✓ However, fine-tuning do worse OOD not just at the end
 - ✓ For model chosen based on OOD test accuracy, LP is still better than FT
2. ID-OOD features get distorted from fine-tuning
 - ✓ For each example x in Living-17, Euclidean distance of features before and after [fine-tuning](#) is calculated
 - ✓ Average distance for ID examples (0.0188 ± 0.0001) is more than for OOD examples (0.0167 ± 0.0001)
 - ✓ For LP-FT, euclidean distance is 20x smaller than for fine-tuning
3. Pretrained features must be good, ID-OOD far apart
 - ✓ Feature Quality
 - Based on MoCo-v1 checkpoint, FT and LP is done on Living-17
 - FT does better than LP
 - ✓ $ID \approx OOD$
 - FT/LP on CIFAR-10 and test on CIFAR-10.1, collected using a similar protocol to CIFAR-10
 - FP outperforms linear probing OOD

Question

4. Conclusion

Summary

- I. Feature distortion generated by fine-tuning underperforms OOD accuracy
- II. Preserving features might be important for robustness
 - ✓ Shown theoretically and empirically in this paper
- III. LP-FT can mitigate tradeoffs between ID and OOD accuracy
- IV. Some limitations remains
 - ✓ Dealing with non-linear activations / different layerwise learning rate / effect of explicit regularization

Thank you